

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

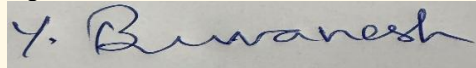
CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	Buvanesh Y	Reg. No.:	192372367
Date:	07/10/2026	Duration:	60 minutes

Code of Conduct

I Buvanesh Y / 192372367 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
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5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.
- Explain how the solution addresses the identified challenges.
- Discuss potential risks, implementation challenges, and limitations.
- Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/Framework	Selects the most suitable	Suitable methodology	Methodology selected but	Outcomes, limitations, or

Justification (10%)	SDLC/model/frame work and provides strong technical justification.	selected with adequate justification.	justification is weak.	future enhancements are missing; report lacks organization and clarity.
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Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

CASE STUDY ANALYSIS

Problem Statement: Online Examination Management System

Problem Statement:

Develop an Online Examination Management System that provides a centralized platform for creating and managing question banks, scheduling examinations, conducting online tests, automatically evaluating objective questions, and generating examination results.

1. Understand the Problem Statement

1.1 Explanation of the Problem

Traditional examination systems generally involve preparing question papers manually, printing them, conducting examinations in physical classrooms, collecting answer sheets, evaluating them, and preparing results. This process requires considerable time, paperwork, human effort, and administrative coordination.

The proposed Online Examination Management System aims to digitalize this complete process. The system provides a common platform where administrators and teachers can create and maintain question banks, prepare examinations, schedule tests, and monitor examinations. Students can log into the system, attend examinations within the specified time, submit their answers, and receive their results.

For objective-type questions such as multiple-choice questions, true/false questions, and similar automatically evaluable questions, the system can compare the student's answer with the correct answer stored in the database and calculate the marks automatically.

Therefore, the system reduces manual work, improves examination efficiency, minimizes evaluation errors, and allows results to be generated quickly.

1.2 Objectives of the Proposed System

The major objectives of the Online Examination Management System are:

1. To digitalize the examination process from question creation to result generation.
2. To create and maintain a centralized question bank for different subjects and topics.
3. To allow teachers to create examinations using questions from the question bank.
4. To schedule examinations with specific dates, starting times, and durations.
5. To allow students to attend examinations online using a secure login.
6. To automatically evaluate objective questions and calculate marks.
7. To generate examination results quickly and accurately.

8. To reduce paperwork and manual effort involved in traditional examinations.
9. To securely store examination and student records in a centralized database.
10. To provide an easy-to-use platform for administrators, teachers, and students.
11. To improve the efficiency and reliability of the examination process.
12. To provide better examination management and monitoring for educational institutions.

1.3 Scope of the Problem

The scope of the system covers the complete management of online objective examinations.

The system includes:

- Administrator login and management.
- Teacher/examiner management.
- Student management.
- Question bank creation.
- Adding, editing, and deleting questions.
- Question categorization based on subject/topic.
- Examination creation.
- Examination scheduling.
- Online examination conduction.
- Timer-based examination.
- Answer submission.
- Automatic evaluation of objective questions.
- Mark calculation.
- Result generation.
- Result viewing.
- Secure storage of examination records.

The system can be used by schools, colleges, universities, coaching centres, training institutions, and organizations conducting online assessments.

The scope can be expanded in the future to include online proctoring, AI-based cheating detection, descriptive-answer evaluation, analytics, and mobile applications.

2. Identify Key Stakeholders and Challenges

The assessment requires identifying the stakeholders, their responsibilities, and the challenges they face. CSA10-CO1 AT1 Case Study Analys...

2.1 Stakeholders

1. Administrator

The administrator is responsible for managing the overall system.

Responsibilities:

- Manage student and teacher accounts.
- Create and manage user roles.
- Manage subjects and examinations.
- Monitor examination activities.
- Maintain system settings.
- Manage question banks.
- Monitor system performance.
- Maintain security and access permissions.

2. Teacher/Examiner

Teachers are responsible for preparing and conducting examinations.

Responsibilities:

- Create questions.
- Add questions to the question bank.
- Edit or remove questions.
- Organize questions according to subjects and topics.
- Create examinations.
- Select questions for an examination.
- Set examination date and duration.
- Monitor examination progress.
- View student performance and results.

3. Student

Students are the primary users who attend the online examination.

Responsibilities:

- Log into the system using valid credentials.
- View available examinations.
- Read examination instructions.
- Attend the examination within the scheduled time.
- Select or enter answers.
- Submit the examination.
- View the generated results.

4. Institution Management

College or institution management may use the system to monitor examination activities and student performance.

Responsibilities:

- Monitor examination statistics.
- Review overall student performance.
- Analyze examination results.
- Ensure proper examination procedures are followed.

5. System/Database Administrator

This stakeholder is responsible for the technical infrastructure.

Responsibilities:

- Maintain the database.
- Manage backups.
- Monitor server performance.
- Maintain system availability.
- Handle technical issues.
- Ensure data security.

2.2 Major Challenges

1. Security

Student information, examination questions, answers, and results must be protected from unauthorized access.

2. Examination Integrity

The system should reduce opportunities for students to access unauthorized resources or share answers during examinations.

3. Large Number of Users

If hundreds or thousands of students attend an examination simultaneously, the system must handle the increased load without crashing.

4. Internet Connectivity

Students require a stable internet connection. Network interruptions may cause problems while attempting or submitting an examination.

5. Question Management

Managing a large number of questions manually can become difficult. Questions need to be categorized and maintained properly.

6. Accurate Evaluation

The system must correctly compare submitted answers with the stored correct answers and calculate marks.

7. Data Loss

Student answers and examination results must not be lost because of server failure or technical problems.

8. Time Management

Examinations must automatically start and end according to the scheduled time and duration.

9. User Errors

Students or teachers may accidentally submit answers, delete questions, or enter incorrect information. The system should provide suitable validation and confirmation.

3. Analyze the Current Approach

3.1 Existing System

In a traditional examination system, the teacher prepares the question paper manually. The question paper is printed and distributed to students in a classroom. Students write their answers on paper and submit their answer sheets after completing the examination. The teacher or examiner then manually evaluates the answer sheets, calculates marks, prepares the result, and publishes it.

Traditional Workflow:

Question Preparation → Printing → Examination → Answer Collection → Manual Evaluation → Mark Calculation → Result Preparation → Result Publication

This process works for conventional examinations but becomes difficult when a large number of students and examinations are involved.

3.2 Strengths of the Existing System

1. Easy to understand and use.
2. Does not require an internet connection.
3. Students can interact directly with teachers.
4. Suitable for institutions with limited technical infrastructure.
5. Can support descriptive answers easily.

3.3 Limitations of the Existing System

1. Time-consuming

Manual question-paper preparation, evaluation, and result generation require significant time.

2. High Paper Usage

Printed question papers and answer sheets require large amounts of paper.

3. Manual Evaluation

Teachers have to check answer sheets individually, which increases workload.

4. Possibility of Human Errors

Mistakes may occur while evaluating answers or calculating total marks.

5. Delayed Results

Results may take several days to prepare, particularly when there are many students.

6. Difficult Record Management

Maintaining physical answer sheets and examination records requires storage space.

7. Difficult to Conduct Large-Scale Examinations

Managing a large number of students manually requires more classrooms, staff, and administrative work.

3.4 Problems/Gaps to Be Addressed

The proposed system addresses the following gaps:

Existing Problem	Proposed Solution
Manual question preparation	Digital question bank
Paper-based examination	Online examination
Manual evaluation	Automatic evaluation
Delayed results	Instant result generation
Physical records	Digital records
Difficult question management	Centralized question bank
Manual mark calculation	Automatic mark calculation
Difficult monitoring	Digital examination monitoring
High administrative workload	Automated examination management

The assessment specifically expects analysis of the existing system, its strengths, limitations, and gaps. CSA10-CO1 AT1 Case Study Analys...

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Proposed Solution

The proposed solution is a web-based Online Examination Management System that provides a centralized platform for managing examinations.

The system will have different modules for administrators, teachers, and students. All important information such as users, questions, examinations, answers, and results will be stored in a centralized database.

Proposed Workflow

Admin/Teacher Login → Create Question Bank → Create Examination → Schedule Examination → Student Login → Attend Examination → Submit Answers → Automatic Evaluation → Result Generation

This solution follows software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security, which are specifically mentioned in the assessment requirements. CSA10-CO1 AT1 Case Study Analys...

4.2 Major Functional Modules

Module 1: User Authentication

This module provides secure login for administrators, teachers, and students.

Functions:

- Login.
- Logout.
- Password management.
- Role-based access.
- User authentication.

For example, a student should only be able to access student-related functions, while a teacher can manage questions and examinations.

Module 2: Question Bank Management

This module allows teachers to create and manage questions.

Functions:

- Add questions.
- Edit questions.
- Delete questions.
- Add multiple-choice options.
- Specify correct answers.
- Categorize questions by subject/topic.
- Store questions for future examinations.

A centralized question bank avoids repeatedly creating the same questions.

Module 3: Examination Management

Teachers can create examinations using questions from the question bank.

Functions:

- Create examination.
- Select questions.
- Set total marks.
- Set examination duration.
- Set examination date.
- Set start and end time.
- Publish or deactivate examination.

Module 4: Student Examination Module

Students can attend examinations through their accounts.

Functions:

- View available examinations.
- Read instructions.
- Start examination.
- Answer questions.
- Navigate between questions.
- View remaining time.
- Submit answers.

The system should automatically submit the examination when the allocated time expires.

Module 5: Automatic Evaluation Module

This is one of the most important modules.

After submission, the system compares each student's answer with the correct answer stored in the question bank.

For example:

Correct Answer: B

Student Answer: B

Result: Correct

Marks: +1

If the student selects the wrong option:

Correct Answer: B

Student Answer: C

Result: Incorrect

Marks: 0

The system then calculates the total score automatically.

Module 6: Result Management

This module generates examination results.

It can display:

- Student name.
- Examination name.
- Total questions.
- Correct answers.
- Incorrect answers.
- Total marks.
- Percentage.
- Result status.

Teachers and administrators can also view the performance of students.

Module 7: Database Management

The database stores:

- Student information.
- Teacher information.
- Questions.
- Correct answers.
- Examination details.
- Student responses.
- Marks.
- Results.

Proper database design helps maintain consistency and prevents unnecessary duplication of data.

4.3 Functional Requirements

The functional requirements describe what the system should do.

1. The system shall allow users to log in securely.
2. The administrator shall be able to manage users.
3. Teachers shall be able to create and manage questions.
4. Teachers shall be able to create question banks.
5. Teachers shall be able to create examinations.
6. Teachers shall be able to schedule examinations.
7. Students shall be able to view available examinations.
8. Students shall be able to attend examinations online.
9. The system shall record student answers.
10. The system shall automatically evaluate objective questions.
11. The system shall calculate examination marks.
12. The system shall generate results.
13. Students shall be able to view their results.
14. Teachers shall be able to view student performance.
15. The system shall securely store examination records.

4.4 Non-Functional Requirements

1. Security

The system must protect user accounts, questions, answers, and results from unauthorized access.

2. Performance

The system should respond quickly, especially when many students are attending an examination simultaneously.

3. Reliability

The system should operate correctly throughout the examination and minimize the possibility of data loss.

4. Usability

The interface should be simple enough for students and teachers to use without extensive training.

5. Scalability

The system should be capable of supporting additional students, teachers, subjects, and examinations in the future.

6. Maintainability

The system should be designed so that individual modules can be modified or upgraded without affecting the entire application.

7. Availability

The system should be available during scheduled examination periods.

4.5 Application of Software Engineering Principles

Modularity

The system is divided into independent modules such as:

Login → Question Bank → Examination → Student Test → Evaluation → Results

This makes development and testing easier.

Scalability

The system should be designed to accommodate increasing numbers of students and examinations.

Maintainability

The code should be properly organized and documented so that future developers can easily modify the system.

Reliability

Answers and examination data should be stored safely. Backup mechanisms can be used to reduce the risk of data loss.

Usability

The interface should provide clear buttons, instructions, timers, and navigation options.

Security

Authentication, authorization, password protection, and controlled access should be implemented to protect sensitive examination information.

These principles correspond directly to the software-engineering principles listed in the assessment rubric.

5. Justify the Chosen Methodology/Framework

Selected Methodology: Agile

The Agile Software Development Methodology is suitable for the Online Examination Management System.

The assessment allows methodologies/frameworks such as Agile, Scrum, Waterfall, Spiral, and DevOps, and requires the selected approach to be justified. CSA10-CO1 AT1 Case Study Analys...

Why Agile is Suitable

The Online Examination Management System consists of multiple modules. Instead of developing the complete system at once, the modules can be developed incrementally.

Agile Development Process

Sprint 1 – User Management

Develop:

- Login.
- Registration/user creation.
- User roles.
- Authentication.

Sprint 2 – Question Bank

Develop:

- Add question.
- Edit question.
- Delete question.
- Categorize questions.

Sprint 3 – Examination Management

Develop:

- Create examination.
- Select questions.
- Set duration.
- Schedule examination.

Sprint 4 – Online Examination

Develop:

- Student examination interface.
- Timer.
- Answer selection.

- Answer submission.

Sprint 5 – Evaluation and Results

Develop:

- Automatic evaluation.
- Mark calculation.
- Result generation.
- Result display.

Sprint 6 – Testing and Security

Perform:

- Functional testing.
- Security testing.
- Performance testing.
- User acceptance testing.
- Bug fixing.

Advantages of Agile for This Project

1. Incremental development – The system can be developed module by module.
2. Early testing – Each module can be tested after development.
3. User feedback – Teachers and students can provide feedback during development.
4. Easy changes – Requirements can be modified when necessary.
5. Reduced risk – Problems can be identified early.
6. Continuous improvement – Each sprint can improve the previous version.
7. Better maintenance – Modules can be updated independently.

Therefore, Agile is suitable because it provides flexibility and supports continuous development and improvement.

6. Discuss Expected Outcomes and Limitations

6.1 Expected Outcomes

The proposed system is expected to provide the following benefits:

1. Reduced Manual Work

Teachers do not need to manually evaluate every objective question.

2. Faster Results

Results can be generated immediately after students submit their examinations.

3. Reduced Paper Usage

The examination can be conducted digitally without printed question papers and answer sheets.

4. Better Question Management

Teachers can maintain a centralized question bank and reuse questions for future examinations.

5. Improved Accuracy

Automatic evaluation reduces errors in calculating objective-question marks.

6. Easy Access

Students can attend examinations from any suitable location with the required internet and system access.

7. Centralized Data

Student, examination, question, answer, and result information can be maintained in one system.

8. Better Monitoring

Teachers and administrators can monitor examinations and analyze student performance.

6.2 How the Solution Addresses Identified Challenges

Challenge	Solution
Manual evaluation	Automatic evaluation
Paperwork	Digital examination
Result delay	Automatic result generation
Question management	Centralized question bank
Unauthorized access	Authentication and authorization
Data loss	Database backup and secure storage
Large number of students	Scalable system architecture
Examination time management	Automatic timer
Human calculation errors	Automated mark calculation

6.3 Potential Risks

1. Internet Failure

Students may lose connectivity during the examination.

Solution: Save answers periodically and allow recovery when the connection is restored.

2. Server Failure

A server problem could interrupt the examination.

Solution: Use reliable servers, regular backups, and recovery mechanisms.

3. Security Threats

Unauthorized users may attempt to access questions or results.

Solution: Use authentication, authorization, encryption, and secure database access.

4. Cheating

Students may attempt to use unauthorized resources.

Solution: Randomize questions/options and, in future versions, implement online proctoring.

5. High System Load

Many students accessing the system simultaneously may affect performance.
Solution: Use scalable infrastructure and performance testing.

6.4 Limitations

Although the proposed system provides many advantages, it has some limitations:

- It depends on internet connectivity.
- Technical problems can affect examinations.
- Initial development requires time and resources.
- Teachers and students may require basic training.
- Objective questions can be evaluated automatically, but descriptive answers require additional evaluation methods.
- Online examinations may require additional security measures to maintain examination integrity.

6.5 Future Enhancements

The system can be enhanced in the future with:

1. AI-based cheating detection.
2. Webcam-based online proctoring.
3. Random question generation.
4. AI-based question generation.
5. Mobile application support.
6. Descriptive-answer evaluation using AI.
7. Student performance analytics.
8. Difficulty-level analysis of questions.
9. Automatic personalized question recommendations.
10. Email/SMS notifications for scheduled examinations.
11. Cloud-based deployment.
12. Advanced dashboards for teachers and administrators.

Conclusion

The Online Examination Management System provides an efficient and centralized solution for managing the complete online examination process. It replaces many time-consuming manual activities with automated digital processes. The system allows teachers to create question banks and examinations, schedule tests, and monitor student performance, while students can attend examinations and receive results through the same platform.

Automatic evaluation of objective questions significantly reduces the workload of teachers and helps generate results quickly. The application of modularity, scalability, maintainability, reliability, usability, and security makes the proposed system more effective and easier to maintain. CSA10-CO1 AT1 Case Study Analys...

The Agile methodology is appropriate because the system can be developed incrementally through separate modules and improved continuously using feedback. Overall, the proposed system can reduce administrative effort, minimize evaluation errors, improve result processing, and provide a more convenient examination experience for both students and teachers.